

SAMBP Technical Report TR-77/2028

PMU-Based Real-Time State Estimation for SAMBP

WLS + EKF, PMU Observability, Bad Data Detection, IBR Parameter Estimation: 35-Bus
Test System

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Contents

1	Background and Objectives	1
1.1	Motivation	1
1.2	Objectives	1
2	Theory and Methodology	1
2.1	WLS State Estimation	1
2.2	Dynamic EKF for IBR Tracking	2
2.3	PMU Observability Analysis	2
2.4	LNR Bad Data Detection	2
3	Simulation Results	2
3.1	PMU Placement Results: 35-Bus System	2
3.2	State Estimation Accuracy	2
3.3	LNR Bad Data Performance	3
4	Conclusion	3

1 Background and Objectives

1.1 Motivation

Power system state estimation provides the real-time network operating point that underpins adaptive protection. With PMU measurements, the state estimation can be updated at synchrophasor reporting rates (50–200 frames/second), enabling dynamic state tracking impossible with SCADA-based SE (2–4 scans/minute). For SAMBP, the key outputs are the IBR parameters k_{ibr} and f_{gfm} , which drive the adaptive relay settings (TR-70 reach adjustment, TR-74 polarisation, TR-57 SPRT threshold) [1].

1.2 Objectives

1. Implement Weighted Least Squares (WLS) state estimation with PMU data.
2. Extend to dynamic EKF for tracking IBR state transitions.
3. Perform PMU observability analysis for 35-bus test system.

4. Implement Largest Normalised Residual (LNR) bad data detection.
5. Achieve estimation error $< 0.5\%$ in normal conditions, $< 2\%$ during fault.

2 Theory and Methodology

2.1 WLS State Estimation

The WLS SE solves the measurement model $\mathbf{z} = h(\mathbf{x}) + \mathbf{e}$, where $\mathbf{z}$ is the PMU measurement vector (voltage phasors at PMU buses, current phasors at branch ends), $\mathbf{x} = [V_i, \theta_i]_{i=1}^n$ is the state vector, and $\mathbf{e} \sim \mathcal{N}(0, \mathbf{R})$ is measurement noise [2].

The normal equations (linearised about current state estimate):

$$\hat{\mathbf{x}} = (\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{R}^{-1} \mathbf{z} \quad (1)$$

For full PMU observability, $\mathbf{H}$ is the full measurement Jacobian and the system is *linear* (no iterative solution required).

2.2 Dynamic EKF for IBR Tracking

The extended state vector augments bus voltages with IBR parameters:

$$\mathbf{x}_{\text{ext}} = [V_i, \theta_i, k_{\text{ibr}}^{(k)}, f_{\text{gfm}}^{(k)}] \quad (2)$$

The process model for k_{ibr} and f_{gfm} uses a random-walk model with process noise $q_{k_{\text{ibr}}} = 0.01 \text{ pu}^2/\text{s}$, appropriate for slow wind/solar variability. During fault conditions, the IBR parameters change rapidly; the EKF innovation gain is increased by a factor of 10 when the residual $\|\mathbf{z} - h(\hat{\mathbf{x}})\|_2 > 3\sigma_z$.

2.3 PMU Observability Analysis

The minimum PMU placement for full observability is found by integer programming:

$$\min \sum_{i=1}^n u_i \quad \text{s.t.} \quad \mathbf{A}\mathbf{u} \geq \mathbf{1} \quad (3)$$

where $\mathbf{A}$ is the bus connectivity matrix augmented for zero-injection buses, and $u_i \in \{0, 1\}$ indicates PMU placement at bus i .

2.4 LNR Bad Data Detection

The Largest Normalised Residual test computes: $r_i^{\text{norm}} = r_i / \sqrt{\Omega_{ii}}$, where $r_i = z_i - h_i(\hat{\mathbf{x}})$ and $\Omega = \mathbf{R} - \mathbf{H}(\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^T$ is the residual covariance. Bad data detected if $\max_i r_i^{\text{norm}} > \chi_\alpha^2$ ($\alpha = 0.05$ significance level; threshold ≈ 3.0 for large systems).

3 Simulation Results

3.1 PMU Placement Results: 35-Bus System

Remark 1. *Two buses (bus 18 and bus 29) achieve only indirect observability through zero-injection constraints. In the event of a PMU failure at an adjacent bus, these buses become unobservable. The placement includes one redundant PMU (bus 7) to cover this contingency.*

Table 1: PMU placement optimisation results for 35-bus test system.

Metric	Value
Total buses	35
PMUs required for full observability	12
Observability with 12 PMUs	95% (33/35 buses)
Zero-injection buses (ZIB) used	4
Redundancy index (avg measurements/bus)	2.3

Table 2: WLS state estimation accuracy: 35-bus system.

Condition	V error (%)	θ error (deg)	k_{ibr} error (%)	f_{gfm} error (%)
Normal (steady-state)	0.12	0.04	0.31	0.28
Fault (within 100 ms)	1.84	0.62	1.95	1.72
Bad data (1 PMU)	0.18	0.06	0.42	0.38

3.2 State Estimation Accuracy

Finding 1 (PMU-based SE meets SAMBP accuracy requirements). *WLS state estimation with 12 PMUs achieves V error $< 0.2\%$ in normal conditions and $< 2\%$ during fault, meeting the SAMBP adaptive protection requirements. IBR parameter estimation (k_{ibr} , f_{gfm}) errors are $< 0.5\%$ in normal and $< 2\%$ during fault.*

3.3 LNR Bad Data Performance

Table 3: LNR bad data detection performance.

Metric	Value
Detection rate (injected bad data)	98.5%
False alarm rate	1.5%
Bad data correction success rate	96.0%

4 Conclusion

Finding 2 (12 PMUs sufficient for SAMBP on 35-bus system). *The integer programming placement yields 12 PMUs achieving 95% observability and meeting SAMBP accuracy requirements ($< 0.5\%$ in normal, $< 2\%$ in fault). LNR bad data detection achieves 98.5% detection rate.*

Finding 3 (EKF IBR tracking enables adaptive protection). *The dynamic EKF extension tracks k_{ibr} and f_{gfm} at synchrophasor rates, providing the real-time IBR parameters required by the TR-70 reach adjustment and TR-74 adaptive polarisation algorithms.*

References

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